

# Symmetric normal boxes: the bare parity identity

Claude, with Daniel Murfet

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## Abstract

Interior normal coordinates range over  $(-1, 1]$  rather than  $(0, 1]$ . For the bare monomial integral the coordinate reflections give the exact identity

$$\int_{(-1, 1]^d} u^h e^{-\beta N u^{2k}} du = \prod_i (1 + (-1)^{h_i}) \int_{(0, 1]^d} u^h e^{-\beta N u^{2k}} du$$

(`monomialSymReal_eq`): each odd exponent kills the integral, each even one doubles it — the paper’s parity factor  $(1 + (-1)^{h_i})$  without the  $\frac{1}{2}$  of an averaged measure. Hence the symmetric-box integral has the mixed-ratio asymptotic with the constant multiplied by the parity product (`monomialSymReal_tendsto`). Zero sorry/axiom.

## 1 The things formalised

```
theorem integrableOn_pi_box_marginal (d S F) (hF : IntegrableOn F (piBox (d + 1) S)) :
  IntegrableOn (fun a => b in piBox d S, F (Fin.cons a b)) S
theorem integral_Ioc_symm_of_parity (g) (h : ) (hg : IntegrableOn g (Ioc (-1) 1))
  (hsym : a, g (-a) = (-1) ^ h * g a) :
  a in Ioc (-1) 1, g a = (1 + (-1) ^ h) * a in Ioc 0 1, g a
def symBox (d : ) : Set (Fin d → ) := piBox d (Ioc (-1) 1)
def monomialSymReal (d) (h k) ( N) : :=
  x in symBox d, ( i, x i ^ h i) * exp (-( * N * i, x i ^ (2 * k i)))
theorem monomialSymReal_eq ( ) : d h k N,
  monomialSymReal d h k N = ( i, (1 + (-1 : ) ^ h i)) * monomialBoxReal d h k N
theorem monomialSymReal_tendsto ... : Tendsto (fun N => monomialSymReal (d + 1) h k N
  / (N ^ (-1) * log N ^ (multCount (ratioExp h k) 1 - 1))) atTop
  ( (( i, (1 + (-1 : ) ^ h i)) * monomialMixedConst h k 1 ))
```

## 2 Mathematics

Induction on the dimension: the Bochner peel (unit 188) exposes the first coordinate on both boxes, the inner symmetric integral is the  $d$ -dimensional one at parameter  $N a^{2k_0}$  (induction hypothesis), and the one-dimensional reflection identity handles the peeled coordinate, whose integrand  $a \mapsto a^{h_0} G(a^{2k_0})$  has parity  $(-1)^{h_0}$ . Integrability of the peeled marginal comes from `Integrable.integral_prod_left` along the same measure-preserving transport.

## 3 Paper-facing meaning

This is the bare version of the paper’s interior-coordinate reduction (parity factors  $\frac{1+(-1)^{h_i+\gamma_i}}{2}$  up to the normalisation of the measure). For a general amplitude the reflections produce the signed

combination  $\sum_{\sigma} \prod_i \sigma_i^{h_i} \eta(\sigma \cdot u)$ , not a scalar factor; an odd exponent then cancels only the corner coefficient, not the whole integral. That reduction is a natural follow-up.

## 4 Lean notes

- The reflection identity is cleanest as a statement about an integrable  $g$  with pointwise parity, proved with `integral_add_adjacent_intervals` and `integral_comp_neg` on interval integrals, converting to `Ioc` integrals with `integral_of_le` at the ends.
- `IntegrableOn.congr_fun` with the peel identity transfers marginal integrability to the rewritten integrand; its pointwise goal is a beta-redex (`beta_reduce`).
- `unitBox d = piBox d (Ioc 0 1)` and `symBox d = piBox d (Ioc (-1) 1)` are `rfl`; state them as `have` equations to `rw` into the peel lemma's shape.